INCH-POUND

A-A-58002
1 JUNE 1992

COMMERCIAL ITEM DESCRIPTION

GENERATOR, PULSE 5 Hz to 50 MHz

The General Services Administration has authorized the use of this Commercial Item Description

Abstract. This Commercial Item Description (CID), describes a Pulse Generator 5 Hz to 50 MHz. This CID is meant as a minimum requirement for the Pulse Generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

Salient characteristics. The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein

1. Classification Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description The equipment shall be Type III, Class 5, Style E and Color R in accordance with MIL-T-28800.

2. Order of precedence In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this CID shall take precedence Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. Safety and environmental The equipment shall meet all safety and environmental requirements as specified in MIL-T-28800 for the classification as stated herein

4. Electrical power sources The equipment shall operate from nominal commercial, military, and shipboard power sources of 110/220 volts AC (±10%) at line frequencies of 45 Hz to 60 Hz. Power consumption shall not exceed 100 watts. Transient state conditions shall meet MIL-T-28800 requirements for the classification specified herein.

5. Temperature requirements

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to Resources and Logistics Services Division, SA-ALC/TIRDM, Kelly AFB, TX 78241-5000.

AMSC N/A FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:58Z
Check the source to verify that this is the current version before use.

a. Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to +50°C.

b. Nonoperating temperature. The equipment shall meet its performance and accuracy requirements in a nonoperating environment of -10°C to +60°C.

6 Mean time between failures (MTBF) The equipment Mean Time Between Failures (MTBF) shall be 2,000 hours The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

7 Pre-conditioning requirement Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase

8 Calibration and maintenance adjustments The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours.

9 Performance characteristics

a Operational modes

(1) Continuous The generator shall oscillate at the selected frequency.

(2) Triggered. The generator shall wait for either a manual activation, via the push of a button, or an electrical signal passing a threshold, before it generates one pulse

(3) Gated The operation shall be identical to that of the triggered mode except that after triggering has occurred the generator shall operate for the duration of the external signal

(4) Double pulse This mode shall be available for continuous, triggered, and gated modes, and shall provide two pulses each period

(5) External width The generator shall be capable of generating a pulse, the width of which is determined by the time an external trigger signal remains above 1 V dc or below -1 V dc

(6) Trigger burst. The Pulse Generator shall be capable of receiving an external trigger that shall trigger a continuous pulse stream. The pulse stream shall start on the rising or falling edge of the external trigger and shall end on the complement edge of the trigger signal.

b. Time domain.

(1) Period range. The Pulse Generator shall have a minimum period range of 200 ms to 20 ns.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

(2) Selector. The frequency shall be selected by a continuously variable control. If range switching is employed, each range shall overlap all adjacent ranges by at least 4 percent, and be continuously adjustable within each range.

(3) Period jitter. The period jitter shall not exceed 0.1 percent +50 ps for any rate setting

(4) Width The pulse width shall be continuously adjustable within a minimum range of 10 ns to 100 ms.

(5) Duty cycle. The maximum duty cycle shall be at least 70% for frequencies to 5 MHz (200 ns), and at least 30% for frequencies from 5 MHz to 50 MHz (200 ns to 20 ns)

(6) Width jitter. The width jitter shall not exceed 0.1 percent +50 ps for any rate setting.

(7) Pulse delay. The pulse generator shall have the capability to delay the occurrence of a pulse between the time range of 35 ns to 100 ms with respect to the synch output.

a. The pulse delay shall be selected by a continuously variable control.

b. Delay jitter The pulse generator's delay jitter, shall be less than 0.1% +50 ps for any setting.

c. Duty cycle The maximum duty cycle shall be at least 70% for frequencies to 5 MHz (200 ns), and at least 30% for frequencies from 5 MHz to 50 MHz (200 ns to 20 ns).

(8) Transition time. The leading and trailing edges of the pulse shall be independently adjustable through the range of 7 ns to 250 ms.

a. The transition time shall be continuously adjustable and shall be split over a minimum of four ranges. Each range shall overlap the adjacent ranges by at least 1%.

b. The control for the transition time shall allow for minimum adjustment of 50:1 on all ranges except for the range which covers 5 ns where a minimum adjustment of 25:1 is acceptable.

c. Linearity. The transition time linearity shall be 5% or less when measured between the 10% and 90% points on the pulse

c. Outputs. The pulse generator shall have the following front panel mounted female BNC outputs. All outputs shall have short protection that shall allow the pulse generator to return to normal operation upon removal of the short.

(1) Source impedance. The source impedance shall be 50 ohms +10% shunted by 35 pF or less.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

(2) Maximum output. The maximum output shall be at least 5 volts into 50 ohms and 10 volts into an open circuit.

(3) Minimum output. The minimum output shall be 1.0 volts into 50 ohms.

(4) Pulse dynamic range The dynamic range shall be at least 10 volts when terminated into 50 ohms

(5) Preshoot, overshoot and ringing The maximum allowable preshoot, overshoot and ringing shall be +(5% of the amplitude setting +100 mV)

(6) Sync pulse The sync pulse output shall be 0 to +1 volt or greater from a 50 ohm source

(7) Normal/complement selection A front panel mounted switch shall be provided to reverse the normally active and quiescent levels in complement form, for all outputs except the sync pulse

d. Input characteristics.

(1) External trigger A front panel mounted female BNC input shall be provided

(2) Minimum amplitude The minimum amplitude that causes a trigger shall be +0 8V p-p pulse 1 7V p-p sine from dc to 50 MHz

(3) Maximum amplitude The maximum level that will cause a trigger shall be at least +7V

(4) Signal width. The signal which causes the trigger shall have a minimum width 10 ns or less

(5) Trigger level The trigger level shall be selectable to occur on either the falling or rising edge.

(6) Manual trigger A front panel mounted push button shall be provided to trigger the pulse generator

e. Settling time. The output signal shall settle to within 2 percent of the final value (both frequency and amplitude) in 0 1 seconds or less for any change or combination of changes in output and input parameters

10 Equipment dimensions The maximum dimensions per component for the pulse generator should be approximately 5.5 inches height x 11 5 inches wide x 11.5 inches deep

11. Equipment weight. The weight including all accessories (excluding manual) per component, shall not exceed 15 lbs.

12. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:58Z
Check the source to verify that this is the current version before use.

conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

13. Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No 376, and all other requirements of this Commercial Item Description are met

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units a request should be made to the contracting officer to determine if the product is acceptable The contracting officer has the option of accepting or rejecting the product.

14. Regulatory requirements In accordance with Section 23.043 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

15 Preservation, packaging, packing, labeling, and marking The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order

Notes

a. MIL-T-28800 is available from the Navy Publishing and Printing Service Office, Fourth Naval District, 700 Robbins Ave , Philadelphia, PA, 19111-5094.

b The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required

Manufacturer	Model	NSN
Hewlett-Packard	8012B	6625-01-005-7144
Wavetek	801	6625-01-071-8403

Custodian:
Air Force - 99

Reviewing activities:
Army - CR
Navy - EC
DLA - ES

Preparing activity:
Air Force - 82

(Project 6625-0691)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.